

Automation Work Overview

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The following projects represent automation systems I personally designed and built for enterprise clients at Begin Software, as well as a prototype built to demonstrate applied AI thinking. Each project replaced a manual, error-prone process with a reliable, scalable system.

Compliance Deadline Automation | *Regulatory Compliance Client*

- Replaced a manual compliance tracking process — when a new entity signs up via an Airtable interface, their data triggers a Make.com automation that calculates all applicable regulatory deadlines
- Deadline logic is dynamic: the automation branches based on company type, state, formation date, and other entity-specific factors to determine which filings apply and when they are due
- Make.com scenarios handle conditional routing — different company types and states follow different compliance calendars, all computed automatically at onboarding
- Resulting deadlines are written back to Airtable, where the client team monitors status, receives alerts, and tracks completion across their entire book of entities
- Impact: Eliminated manual deadline research and calendar-building per new client — a process that previously took hours per entity now completes in seconds at signup

Stack: Make.com · Airtable

TesTank — Field Test & Certification System | *Equipment Testing Client*

- Built a full job scheduling and test management platform for field technicians testing tanks, lines, and manometers
- Technicians submit structured test results in the field; system auto-generates compliance certificates upon passing
- Complex SQL with UNION queries across 3 equipment tables, versioning logic, and duplicate validation
- Auto-incremented naming conventions (T1, T2...) and calibration date calculations handled in JavaScript changesets
- Impact: Replaced a multi-step manual process — test, record, email certificate — with a single-screen workflow

Stack: Retool · Xano · PostgreSQL · JavaScript

AI-Powered BPMN Workflow Builder | *Internal Tool — Begin Software*

- Multi-agent AI pipeline that converts plain-language meeting transcripts into interactive BPMN process diagrams
- Agent 1 handles branching workflow logic; Agent 2 generates BPMN XML; Agent 3 computes DI layout coordinates
- Custom DFS-based diagram renderer built in Softr renders valid BPMN XML to interactive SVG in the browser
- Presented live to a mixed client/internal audience; delivered under budget with a decision-tree AI feature added
- Impact: Turned a 2–3 hour manual diagramming exercise into a 2-minute automated output

Stack: Softr · Xano · Make.com · Camunda · Claude API · JavaScript

Plasma Donor Eligibility Screener | *Prototype — Parachute Plasma Donation Centers*

- Identified a real operational pain point: donation center staff spend significant time fielding eligibility questions from donors who may not qualify — a manual, inconsistent, and labor-intensive screening process
- Built a conversational AI agent powered by the Claude API that walks potential donors through FDA eligibility criteria in natural language — replacing a staff-handled Q&A with a self-serve pre-screening tool
- Full eligibility ruleset (age, weight, medications, travel history, tattoos, recent illness, surgery, pregnancy) encoded as a structured system prompt; agent dynamically branches based on donor responses
- Returns a clear Eligible / Deferred / Call Center verdict with reasoning and, if eligible, surfaces a direct booking CTA to the Parachute app
- Built end-to-end in approximately 4 hours using Cursor and Claude as coding agents — scaffolding, API wiring, and edge case debugging all accelerated through agentic pair programming
- Impact: Demonstrates how a high-volume manual intake process can be automated with an AI agent while maintaining accuracy, regulatory compliance, and a branded user experience

Stack: React · Vite · Claude API · Cursor

Cross-Project Patterns

Tooling	Make.com, Airtable, Retool, Xano, Softr, React — selected based on client needs and use case
AI Integration	Claude API for multi-agent orchestration and conversational agents; agents handle specialization (logic, XML, layout, eligibility reasoning)
QA Approach	Console.log-heavy debug cycles, server-side validation, real client data testing, and manual scenario testing before handoff
Role-Based Access	Permission groups with Retool groups API; different views and write access per user type
Coding Agents	Used Claude and Cursor throughout — scaffolding, debugging complex SQL, iterating on JS logic, and building the Parachute prototype end-to-end

Note: Begin Software client work performed under NDA. Project names and specific client details have been generalized. Parachute screener is an independent prototype built to demonstrate applied AI workflow thinking.